

A no-go theorem for the literal unbounded-operator extension

Fractional difference equations with finite delay

Candidate partial-result packet

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Status. Candidate sharp obstruction, likely valid pending expert review.

1 Source direction

Zhang and Bu study, for $2 < \alpha < 3$, the explicit fractional difference equation

$$\Delta^\alpha u(n) = Au(n) + \gamma u(n - \lambda) + f(n), \quad n \in \mathbb{N}_0, \quad (1)$$

with

$$u(-\lambda) = \cdots = u(0) = u(1) = u(2) = 0. \quad (2)$$

Their coefficient A is bounded. The introduction says that systems including unbounded operators remain an open problem [1]. The source statement is a broad research direction rather than a quantified conjecture.

We notice that the fractional difference equation (1.1) in the case $2 < \alpha < 3$ and $\gamma = 0$ was previously studied in [21]. Our results recover the explicit formula of solution and the characterization of ℓ^p -maximal regularity for (1.1) obtained in [21]. The main tool in this paper is the operator-valued Fourier multipliers theorems on $\ell^p(\mathbb{Z}; X)$ obtained by Blunck [5], we will transform the ℓ^p -maximal regularity of (1.1) to an operator-valued Fourier multiplier problem on $\ell^p(\mathbb{Z}; X)$.

We notice that the research in this paper was motivated by recent studies on discrete mathematical models proved that they can serve as a new microstructural basis for fractional nonlocal continuum mechanics and physics [18, 19]. Fractional difference equations can be also used to formulate adequate models in nanomechanics [19, 20] and therefore further studies in this class of fractional difference equations deserve to be investigated. Our contribution in this paper provides a new qualitative advance in this line of research, that incorporates tools from operator theory and allows the analysis of maximal regularity for a very general but still simple model. The study of more complex dynamical systems that include unbounded operators is still an open problem. This task will be the objective of forthcoming works.

This paper is organized as follows: in section 2, we recall some basic concepts and results in the existing literature related to fractional difference operators, UMD spaces, R -boundedness, the discrete time Fourier transform

Figure 1: The unbounded-operator direction in the source introduction, PDF page 3.

This packet tests the closest literal extension: let $A : \mathcal{D}(A) \subseteq X \rightarrow X$ be closed, keep (1)–(2), and ask for a classical solution for every finitely supported X -valued forcing. A classical solution must satisfy $u(n) \in \mathcal{D}(A)$ whenever $Au(n)$ occurs. This is the solution class needed by the usual maximal regularity requirement $Au \in \ell^p(\mathbb{N}_0; X)$.

2 Proof intuition

The scheme is explicit: the third forward difference at time zero determines the value three time levels ahead. Because the first three values of u are zero, an impulse $f(0) = x$ passes through without smoothing and gives $u(3) = x$. Three steps later, the same equation asks to apply A to $u(3)$. Thus every admissible impulse must already lie in $\mathcal{D}(A)$. If all X -valued impulses are allowed, then $\mathcal{D}(A) = X$, and closedness makes A bounded.

This same obstruction is encoded in the source's resolvent sequence: $S_\alpha(0) = I$, so the first recurrence step contains $AI = A$. Retaining a bounded-operator-valued resolvent sequence also forces boundedness. Finally, a diagonal example shows that the frequency-domain resolvent bounds alone cannot restore classical solvability.

3 The domain obstruction

The source defines

$$\Delta^\alpha = \Delta^3 \Delta^{-(3-\alpha)}, \quad \Delta^{-(3-\alpha)} u = k^{3-\alpha} * u,$$

where $k^\beta(0) = 1$ and $\Delta v(n) = v(n+1) - v(n)$.

Theorem 1 (All-data rigidity). *Let X be a Banach space, $2 < \alpha < 3$, $\lambda \in \mathbb{N}$, and $\gamma \in \mathbb{C}$. Let $A : \mathcal{D}(A) \subseteq X \rightarrow X$ be a linear operator. Suppose that for every finitely supported $f : \mathbb{N}_0 \rightarrow X$ there is a classical solution of (1)–(2) on all of $\mathbb{N}_0$. Then $\mathcal{D}(A) = X$. In particular, if A is closed, then $A \in \mathcal{B}(X)$.*

Proof. Fix an arbitrary $x \in X$ and take the impulse forcing

$$f(0) = x, \quad f(n) = 0 \quad (n \geq 1).$$

Set $v = \Delta^{-(3-\alpha)} u = k^{3-\alpha} * u$. From (2),

$$v(0) = v(1) = v(2) = 0, \quad v(3) = k^{3-\alpha}(0)u(3) = u(3).$$

Consequently,

$$\Delta^\alpha u(0) = \Delta^3 v(0) = v(3) - 3v(2) + 3v(1) - v(0) = u(3).$$

Evaluating (1) at $n = 0$ and using the zero initial history gives

$$u(3) = Au(0) + \gamma u(-\lambda) + f(0) = x.$$

Since the solution is classical on all of $\mathbb{N}_0$, equation (1) at $n = 3$ requires $u(3) \in \mathcal{D}(A)$. Hence $x \in \mathcal{D}(A)$. The vector $x \in X$ was arbitrary, so $\mathcal{D}(A) = X$. A closed everywhere defined linear operator on a Banach space is bounded by the closed graph theorem. $\square$

Proposition 1 (The source resolvent sequence is rigid). *Suppose one keeps Definition 3.1 of [1]: all $S_\alpha(n)$ are bounded operators, $S_\alpha(0) = S_\alpha(1) = S_\alpha(2) = I$, $S_\alpha(-j) = 0$ for $1 \leq j \leq \lambda$, and the displayed recurrence is an identity in $\mathcal{B}(X)$. Then its coefficient A is bounded. Thus this definition has no genuinely unbounded instance.*

Proof. At $n = 0$,

$$(k^{\alpha-2} * S_\alpha)(0) = S_\alpha(0) = I, \quad (k^{\alpha-2} * S_\alpha^\lambda)(0) = S_\alpha(-\lambda) = 0.$$

The recurrence therefore expresses A as $S_\alpha(3)$ plus a scalar multiple of I . Both are bounded. $\square$

4 The frequency bounds do not rescue the extension

Write

$$g(z) = z^{3-\alpha}(z-1)^\alpha, \quad q(z) = g(z) - \gamma z^{-\lambda} \quad (|z| = 1).$$

The source's characterization uses the two families

$$\{g(z)(q(z) - A)^{-1}\}, \quad \{z^{-\lambda}(q(z) - A)^{-1}\}, \quad (3)$$

away from the exceptional points of the circle.

Example 1 (Frequency conditions without classical solvability). *Let $H = \ell^2(\mathbb{N}; \mathbb{C})$, put $M = 2^\alpha + |\gamma|$, and define*

$$Ae_j = -(M + 2 + j)e_j, \quad \mathcal{D}(A) = \left\{ x = (x_j) : \sum_{j \geq 1} (M + 2 + j)^2 |x_j|^2 < \infty \right\}.$$

Then A is densely defined, self-adjoint, closed, and unbounded. Both families in (3) are R -bounded, but (1)–(2) is not classically solvable for every finitely supported H -valued forcing.

Proof. For $|z| = 1$,

$$|q(z)| \leq |z - 1|^\alpha + |\gamma| \leq 2^\alpha + |\gamma| = M.$$

Since $\sigma(A) = \{-(M + 2 + j) : j \geq 1\}$,

$$\text{dist}(q(z), \sigma(A)) \geq 3, \quad \|(q(z) - A)^{-1}\| \leq \frac{1}{3}.$$

Also $|g(z)| \leq 2^\alpha$. Hence both families in (3) are uniformly norm bounded. On a Hilbert space, uniform norm boundedness is equivalent to R -boundedness.

Now take $x = (1/j)_{j \geq 1}$. Then $x \in H$ but $x \notin \mathcal{D}(A)$. The impulse $f(0) = x$, $f(n) = 0$ for $n \geq 1$, belongs to every $\ell^p(\mathbb{N}_0; H)$. The proof of Theorem 1 forces $u(3) = x$, which is outside $\mathcal{D}(A)$, so there is no classical solution on all of $\mathbb{N}_0$. $\square$

5 Consequences and limitations

There are two exhaustive outcomes for a naive replacement of $A \in \mathcal{B}(X)$ by a closed operator while retaining the source hypotheses:

- If the bounded source-style resolvent sequence is retained, then Proposition 1 forces A to be bounded, so the extension has no new operators.
- If that hypothesis is dropped and only the frequency families are retained, Example 1 shows that they do not imply all-data classical solvability.

Allowing a mild equation in an extrapolation space does not restore the standard conclusion $Au \in \ell^p(\mathbb{N}_0; X)$ for arbitrary X -valued data: the impulse still gives $u(3) = x$, and $Au(3) \in X$ requires $x \in \mathcal{D}(A)$.

This is a complete no-go result for the literal extension, but only a partial answer to the source's broad research direction. A positive unbounded theory must materially reformulate the problem—for example by imposing compatible data in a domain scale, weakening the target regularity, or using an implicit scheme where a resolvent of A smooths the newest time level.

6 Verification and novelty status

The time-zero identity was independently derived from the source definition and cross-checked against the same computation in its proof of Theorem 3.8. The diagonal example was checked by its exact spectrum and the elementary symbol bound. No numerical evidence is used.

On August 11, 2026, the local result indexes and bounded exact-title, quoted-phrases, and core-term web searches found no later resolution of this obstruction. Novelty confidence is moderate pending a specialist citation search.

References

- [1] J. Zhang and S. Bu, *Maximal regularity for fractional difference equations with finite delay on UMD spaces*, arXiv:2406.15417 (2024).
- [2] S. Blunck, *Maximal regularity of discrete and continuous time evolution equations*, *Studia Math.* **146** (2001), 157–176.